

THE CUSTOMER'S PAIN

You see a charge you never made. Then you typically wait 37 hours to hear back.

A customer who reports a charge they do not recognize waits a **median of 37 hours** for the bank's first response. One in ten waits **58 hours or more**. Until then, the charge sits on their account with no answer.

37 h

median wait for a first response to "Cargo no reconocido" (unrecognized charge)

MEASURED

p90 58 h

n = 7,567 of 12,297

Why disputes: a dispute can be checked in code against the customer's own ledger and decided by an explicit policy.

DESIGN ARGUMENT

Source: docs/analysis/demand-report.md (finding 3: median 37.0 h, p90 58.0 h over the 7,567 of 12,297 complaints with a recorded first response). Measured on the challenge's synthetic LATAM Bank dataset, not a live bank.

A dispute agent that knows when to act, when to ask, and when to step aside.

1 Resolves, with proof

RESUELTO

✓ **Identidad verificada**
Sesión de demostración activa

✓ **Transacción localizada**
Cargo identificado: Uber · COP 38.500 · 14 de jun de 2026

✓ **Política evaluada**
Resolución automática aplicada

✓ Referencia REF-52093EF1A2

The record backs the claim, so a provisional credit lands in seconds, in the same chat.

2 Asks when it is unclear

Debajo encontrará 2 cargos de su cuenta que coinciden con lo que nos contó. Por favor, toque el que no reconoce o seleccione "No está en la lista" si no aparece allí.

Taxi Seguro
COP 27.000
15 de jun de 2026 · Transporte

Taxi Seguro
COP 27.000
15 de jun de 2026 · Transporte

No está en la lista

Escriba su mensaje...

Two matching charges: the customer picks from their own. It never guesses.

3 Protects, hands off

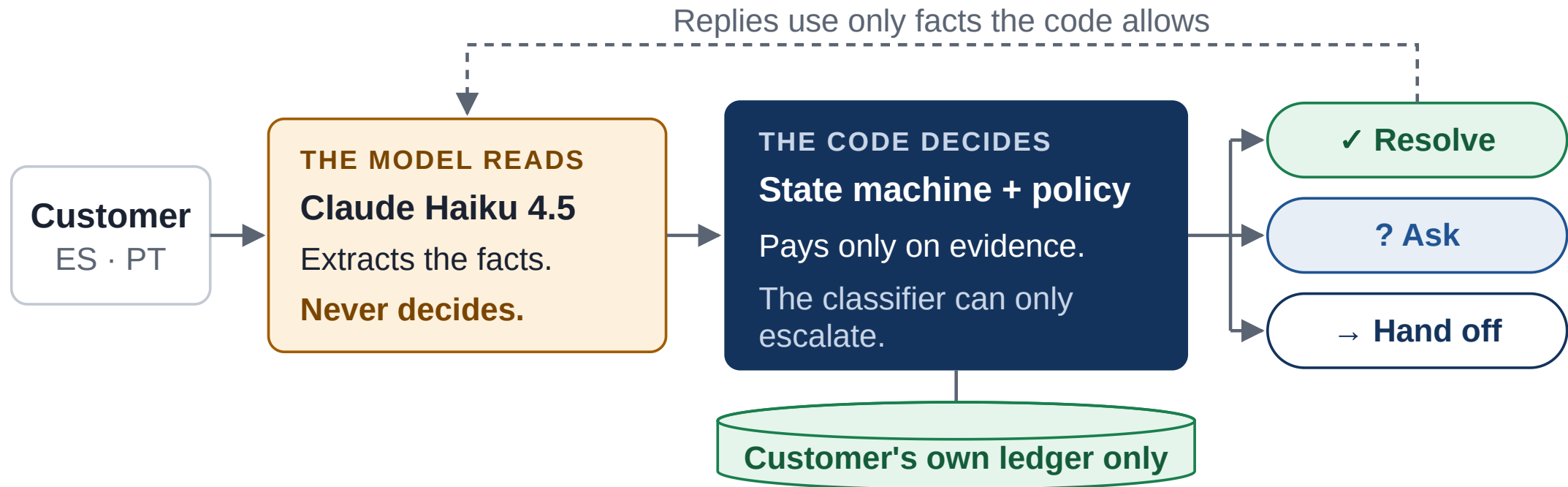
HECHOS VERIFICADOS **Del registro**

Transacción	SYN-DEMO-ONLINE
Comercio	Tienda Online Global
Monto	COP 689.000
Fecha	12 de jun de 2026
Canal	Web

Likely fraud: the card is blocked, and a person gets the verified facts and the customer's account.

Spanish and Portuguese Credits and card blocks are simulated

The model reads. The code decides.



The LLM only extracts and phrases
No refund comes from the model.

Policy and permissions live in code
No lookup can reach another customer's ledger.

Verified facts only
A persuasive claim the record contradicts still goes to a person.

Design argument, enforced in code: an exhaustive sweep over every reason and all 128 combinations of the other conditions shows the classifier never causes a credit (tests/test_policy_not_overridden.py) · 1,121 automated tests · docs/architecture-decisions.md AD-3, AD-5, AD-13 to AD-15.

Honest models: we ship only what the numbers back.

ETL with contracts

CONTRACTS
9 tables, one schema contract each

QUALITY CHECKS
Row counts, duplicate keys, null rates, foreign keys, and a lineage manifest

FOREIGN KEYS **MEASURED**
✓ 0 orphaned rows

Why a subset

TRANSACTIONS **MEASURED**
130,690 of 5M rows: a 30-day window, May 18 to Jun 17, 2026

COMPLAINTS **MEASURED**
67,095, in full

WHY **DESIGN ARGUMENT**
No S3 at runtime (privacy), reproducible, free-plan deploy; a dispute only needs the recent ledger

Fraud model vs the bank

TRAINED OFFLINE **MEASURED**
2,951,642 transactions over 2 years

TEST PR-AUC **MEASURED**

Bank	0.720
Ours	0.707

SHIPPED INSTEAD
A cost-justified gate, **fraud score > 30**. The model only informs the advisor.

Our model lost to the bank's score, so it never decides: the rule the numbers back does.

Sources: etl/schema_contract.py, data/lineage_manifest.json, data/extraction_manifest.json; docs/ml/fraud-model.md, AD-15 (on test, the gate's 48 escalations are all fraud). The priority classifier shows no measured lift (p = 0.45) and can only escalate.

48 held-out conversations on the real model. Zero unsafe outcomes.

0 / 48

unsafe, in each of 3 runs

MEASURED

18 / 18 resolvable cases resolved

26 / 26 card blocks right

3.0 s median reply (p95 7.7 s)

\$0.004 per case

SAME 48 CASES	MEASURED	CORRECT OF 48	UNSAFE OF 48	UNNEEDED TRANSFERS OF 18
Our hybrid		43	0	0
Claude Haiku 4.5, 3 runs				
Regex extractor		43	0	0
model swapped for rules, 1 run				
Always escalate		n/a	0	18
the safety anchor				

Escalating everything is safe but helps no one. The code, not the model, keeps it safe: the 5 misses moved no money. Offline suite, mocked model: dropping the evidence check pays 6 unsafe credits. SIMULATED

Read with its limits: one customer's account, messages written by a model that never saw the policy, labels fixed before the first run. 0 of 48 bounds the unsafe rate below about 7% (95%), not at zero.

ROUTE TO PRODUCTION

From demo to the bank's front line.

STEP 1

Real identity

The bank's identity verification replaces the demo login.

STEP 2

Managed database

Replicated storage across several instances, not local SQLite.

STEP 3

Monitoring

The structured, correlated event log feeds alerting.

STEP 4

Real traffic

Thresholds and the live model re-validated on real customer traffic.

LIVE DEMO

factored-hackaton-latest.onrender.com

CODE

github.com/jAgusGelos/factored-hackathon-2026-jagusgelos

An answer in seconds, not 37 hours. Every refund backed by evidence. Every hard case in a person's hands, with the file already complete.